

Nikita Pavlichenko

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WORK EXPERIENCE

JetBrains AI

Senior Machine Learning Engineer

March 2024 – Present

Berlin, Germany

- Developing LLMs for Cloud Code Completion in JetBrains AI.
- Technical lead for the LLM training project, responsible for data processing, pre-training, fine-tuning, and alignment.
- Trained and deployed LLMs for Code Completion, resulting in a 2x improvement in metrics compared to the baseline OpenAI-based solution. These models are deployed for use in JetBrains' high-selling IDEs, serving millions of users worldwide, and are a core feature of the JetBrains AI product, contributing over \$1 million in revenue.

Yandex LLC

AI Research Scientist

September 2020 – March 2024

Belgrade, Serbia

Research Contributions:

- Lead deep learning research for [Toloka AI](#), resulting in 6 publications on Natural Language Processing, Computer Vision, and generative AI at NeurIPS, SIGIR, and HCOMP.
- Authored the [ICML 2023 tutorial](#) grounded in my original research on RLHF.

Engineering and Development:

- Lead alignment (SFT and RLHF) in Toloka's LLM training project: responsible for dataset collection and trained a model that surpassed Falcon 40B Instruct with >65% win rate on real user requests.
- Develop and maintain an AI-assisted annotation service that showed up to a 30% increase in revenue for text classification projects.
- Develop and maintain the [Crowd-Kit](#) Python package that implements Bayesian models for truth inference (150 GitHub stars, 11 citations).
- Authored 3 patent applications.
- Mentored 2 interns and a junior that was promoted to a mid-level position, and conducted more than 120 interviews (more than 40 hired).

MIPT and Inria NANO/D

Research Intern

March 2020 – September 2020

Moscow, Russia / Grenoble, France

- Developed and implemented the [S-GCN](#) (deep learning model operating on molecular graphs) method for protein model quality prediction that achieved state-of-the-art on CASP MQA challenge.
- Authored a paper on S-GCN accepted to a high-impact journal.

Yandex LLC

Software Engineering Intern / Machine Learning Engineer Intern

July 2018 – November 2018 / July 2019 – October 2019

Moscow, Russia

- Developed and implemented a machine learning model for ranking video clips in web crawler schedules. The model allowed to remove more than 90 % of deleted videos from search results.
- Enhanced the performance speed of all search anti-spam systems by 5% by introducing the fastest C++ hash-table implementation in Yandex.

EDUCATION

SKILLS

Languages: Python, C++, Rust, R, Bash. **Frameworks & Libraries:** PyTorch, Tensorflow, Keras, XGBoost CatBoost, Sci-Kit Learn, Numpy, Pandas, PyTorch Geometric. **Tools & Technologies:** Hadoop, Hive, Spark, AWS, Kubernetes, Docker, Git, Linux, Algorithms, Data Structures. **Domains:** Crowdsourcing, NLP, CV, RL, Graph Machine Learning

PUBLICATIONS

- [1] **Pavlichenko, N.**, Stelmakh, I., & Ustalov, D. (2021). CrowdSpeech and Vox DIY: Benchmark Dataset for Crowdsourced Audio Transcription. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 1.
- [2] Igashov, I., **Pavlichenko, N.**, & Grudinin, S. (2021). Spherical convolutions on molecular graphs for protein model quality assessment. *Machine Learning: Science and Technology*, 2(4), 045005.
- [3] **Pavlichenko, N.**, & Ustalov, D. (2023). Best Prompts for Text-to-Image Models and How to Find Them. *SIGIR '23: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2067–2071.
- [4] Ustalov, D., **Pavlichenko, N.**, Losev, V., Giliyev, I., & Tulin, E. (2021). A general-purpose crowdsourcing computational quality control toolkit for python. *The Ninth AAAI Conference on Human Computation and Crowdsourcing: Works-in-Progress and Demonstration Track (HCOMP 2021)*.
- [5] **Pavlichenko, N.**, & Ustalov, D. (2021). IMDB-WIKI-SbS: An evaluation dataset for crowdsourced pairwise comparisons. *NeurIPS 2021 Data-Centric AI Workshop*.
- [6] Tseytlin, B., **Pavlichenko, N.**, Likhobaba, D., Smirnova, A., & Ustalov, D. *Toloka at LMNL Challenge: Learning from Noisy Labels with Annotator Reliabilities and Self-Supervised Training*.
- [7] Ustalov, D., Fedorova, N., & **Pavlichenko, N.** (2022). Improving Recommender Systems with Human-in-the-Loop. *Proceedings of the 16th ACM Conference on Recommender Systems*, 708–709.
Ustalov, D., Pavlichenko, N., Likhobaba, D., & Smirnova, A. (2023). *WSDM Cup 2023 Challenge on Visual Question Answering*.
- [8] Ustalov, D., **Pavlichenko, N.**, Stelmakh, I., & Kuznetsov, D. (2021). VLDB 2021 Crowd Science Challenge on Aggregating Crowdsourced Audio Transcriptions. *Proceedings of the 2nd Crowd Science Workshop: Trust, Ethics, and Excellence in Crowdsourced Data Management at Scale*, 1–7.
- [9] Ustalov, D., **Pavlichenko, N.**, & Tseytlin, B. (2021). Learning from Crowds with Crowd-Kit. *ArXiv Preprint ArXiv:2109.08584*.
- [10] Ustalov, D., **Pavlichenko, N.**, Tseytlin, B., Baidakova, D., & Drutsa, A. (2022). Web Engineering with Human-in-the-Loop. *International Conference on Web Engineering*, 505–508.

[11] Ustalov, D., Smirnova, A., Fedorova, N., & **Pavlichenko, N.** (2023). Crowdsourcing for Information Retrieval. *European Conference on Information Retrieval*, 357–361.